
EE1060: Differential Equations and Transform Techniques

Lecture 32: Laplace Transform

Topics :

1. Properties of Laplace Transform
 2. Poles and Zeros
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One-Sided Signals and their Laplace Transform

from Lec 31:- $u(t)$ & $-u(-t)$ have the same functional form of LT but have different ROC.

$$\text{ROC of } \mathcal{L}\{u(t)\} : \text{Re}(s) > 0$$

$$\text{ROC of } \mathcal{L}\{u(-t)\} : \text{Re}(s) < 0.$$

for signals def for $t > 0$ (one-sided signals) (or) signals that correspond to $\underline{h(t)}$ of impulse response of a Causal system

$$\text{one-sided Laplace Transform} \quad x(s) = \int_0^{\infty} x(t) e^{-st} dt$$

$$\text{Typical Rep of } x(s) = \frac{N(s)}{D(s)} \rightarrow$$

Roots of $N(s) \rightarrow \text{Zero's}$
Roots of $D(s) \rightarrow \text{Pole's.}$

$$\text{ROC of } e^{-at} u(t) \rightarrow \text{Re}\{s\} > -a.$$

$$x(s) = \frac{1}{s+a} \quad \text{Pole} = -a$$

$$\text{ROC of } -u(-t) \rightarrow \text{Re}\{s\} < 0$$

$$x(s) = \frac{1}{s} \quad \text{Pole: } 0$$

Properties of Laplace Transform

a) Scaling:-
$$\begin{aligned} \mathcal{L}\{c x(t)\} &= \int_0^{\infty} c x(t) e^{-st} dt \\ &= c \int_0^{\infty} x(t) e^{-st} dt \\ &= c X(s) \end{aligned}$$

Roc is unchanged.

b) Linear Combination:-
$$\begin{aligned} \mathcal{L}\{x_1(t) + x_2(t)\} &= \int_0^{\infty} (x_1(t) + x_2(t)) e^{-st} dt \\ &= \int_0^{\infty} x_1(t) e^{-st} dt + \int_0^{\infty} x_2(t) e^{-st} dt \\ &= X_1(s) + X_2(s) \end{aligned}$$

Roc: $\text{Roc}\{X_1(s)\} \cap \text{Roc}\{X_2(s)\}$.

c) Frequency Shift $\mathcal{L}\{e^{at} f(t)\}$

$$\begin{aligned} \mathcal{L}\{e^{at} f(t)\} &= \int_0^{\infty} e^{at} f(t) e^{-st} dt = \int_0^{\infty} f(t) e^{-(s-a)t} dt \\ &= F(s-a) \end{aligned}$$

Roc Shifts

Example:- if $h(t) = u(t)$, then $\lim_{t \rightarrow \infty} h(t) = 1$ (unstable); if $h(t) = e^{-at} u(t)$, $\lim_{t \rightarrow \infty} h(t) = 0$ (stable)

Properties of Laplace Transform

d) Time Shift

$$\mathcal{L}\{f(t-\tau)\} = \int_0^{\infty} f(t-\tau) e^{-st} dt$$

$$= \int_{-\tau}^{\infty} f(t_1) e^{-s(t_1+\tau)} dt_1 = e^{-s\tau} \int_{-\tau}^{\infty} f(t_1) e^{-st_1} dt_1$$

$$\text{as long as } \tau : f(t-\tau) \text{ is one sided} = e^{-s\tau} \int_0^{\infty} f(t_1) e^{-st_1} dt_1 = \underline{\underline{e^{-s\tau} F(s)}}$$

e) Scaling in Time

$$\mathcal{L}\{f(at)\} = \int_0^{\infty} f(at) e^{-st} dt \quad a > 0, \quad = \int_0^{\infty} f(t_1) e^{-s \frac{t_1}{a}} \frac{dt_1}{a}$$

$$= \frac{1}{a} F\left(\frac{s}{a}\right)$$

→ f) Differentiation

$$\mathcal{L}\{f'(t)\} = \int_0^{\infty} f'(t) e^{-st} dt$$

$$= \underbrace{e^{-st} f(t)}_0^{\infty} + s \int_0^{\infty} f(t) e^{-st} dt$$

$$= sF(s) + \left[\lim_{t \rightarrow \infty} \underbrace{e^{-st} f(t)}_{f'0} - f(0) \right] \xrightarrow{\text{True if}} \lim_{t \rightarrow \infty} \frac{f(t)}{e^{mt}} = 0$$

$$= sF(s) - f(0)$$

for some m.

$$\text{In general } \mathcal{L}\{f^{(n)}(t)\} = s^n F(s) - s^{n-1} f(0) - s^{n-2} f'(0) - \dots - f^{(n-1)}(0)$$

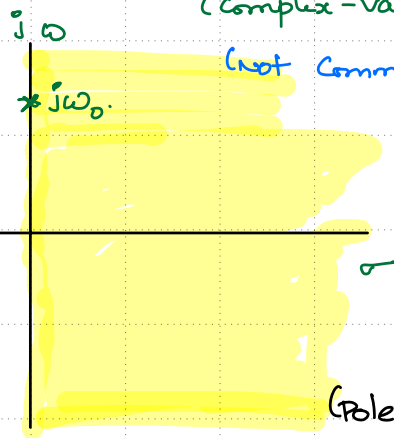
which defines
ROC.

$$\text{Integration: } \mathcal{L}\left\{\int_0^t f(\tau) d\tau\right\} = \frac{F(s)}{s}$$

Examples

Example 1:- if $x(t) = e^{j\omega_0 t} u(t)$
(Complex-valued signal)

(Not Common when dealing with RT Systems)



(Pole-Zero Plot)

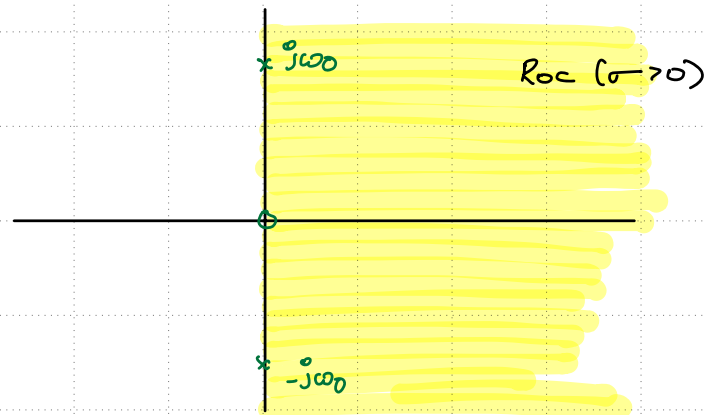
$$\mathcal{L}\{x(t)\} = \int_0^{\infty} e^{j\omega_0 t} e^{-st} dt$$

$$X(s) = \frac{1}{s - j\omega_0} \quad \text{Roc: } \sigma > 0$$

for $x(s) \rightarrow$ Single pole at $s = j\omega_0$

if it is an impulse response of a system $\rightarrow$ marginally stable

Example 2:- if $x(t) = \cos \omega_0 t u(t)$



Roc ($\sigma > 0$)

$$X(s) = \int_0^{\infty} \cos \omega_0 t e^{-st} dt$$

$$= \frac{s}{s^2 + \omega_0^2} \quad (\text{by linearity}) \quad \text{Roc: } \sigma > 0$$

$X(s) \rightarrow$ one zero at $s = 0$

(also considered as two zeros one at $s = 0$ and other at ∞)

Two poles at $s = \pm j\omega_0$.

Example 3:- if $x(t) = \sin \omega_0 t u(t)$

$$X(s) = \frac{\omega_0}{s^2 + \omega_0^2}$$